Scaling invariance of the angular lower-tail coefficient

Statement of the problem

Let C be a copula on $[0,1]^2$. For $\gamma > 0$ and $\delta \in (0,2)$, define

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u},$$

whenever the limit exists.

We prove that, under suitable regularity conditions, $\kappa_L(\delta, \gamma; C)$ does not depend on the choice of the scaling parameter γ. We then verify that the same conclusion applies to the proposed copula C_{PM}.

Assumptions

Fix $\delta \in (0,2)$. Let $\tilde{\gamma} > 0$ be an arbitrary reference scaling parameter. We assume:

(A1) The quantity

$$\tilde{\kappa}(d) := \kappa_L(d, \tilde{\gamma}; C) = \lim_{u \downarrow 0} \frac{\log C\big(\tilde{\gamma} u^{d}, \tilde{\gamma}^{-1} u^{2-d}\big)}{\log u}$$

exists for d in a neighborhood of the fixed point δ, and in particular $\tilde{\kappa}(\delta)$ is finite.

(A2) The convergence above is locally uniform in d near δ: for every $\varepsilon > 0$, there exist $r = r(\varepsilon) > 0$ and $u_0 = u_0(\varepsilon) \in (0,1)$ such that

$$\left| \frac{\log C\big(\tilde{\gamma} u^{d}, \tilde{\gamma}^{-1} u^{2-d}\big)}{\log u} - \tilde{\kappa}(d) \right| \leq \varepsilon$$

for all $u \in (0, u_0]$ and all $d \in (0,2)$ satisfying $|d - \delta| \leq r$.

(A3) The map $d \mapsto \tilde{\kappa}(d)$ is continuous at $d = \delta$.

These are the regularity conditions needed for the ε-argument below.

Notation

For $\gamma > 0$, $d \in (0, 2)$, and $u \in (0, 1)$, define

$$E_\gamma(u; d) := \frac{\log C\big(\gamma u^d, \gamma^{-1} u^{2-d}\big)}{\log u}.$$

Then, by assumption,

$$\lim_{u \downarrow 0} E_{\tilde{\gamma}}(u; d) = \tilde{\kappa}(d)$$

for all d in a neighborhood of δ.

Main theorem

Theorem 1. *Assume* (A1)–(A3). *Then, for every* $\gamma > 0$,

$$\kappa_L(\delta, \gamma; C) = \kappa_L(\delta, \tilde{\gamma}; C) = \tilde{\kappa}(\delta).$$

Hence, for the fixed $\delta \in (0, 2)$, *the value of* $\kappa_L(\delta, \gamma; C)$ *is independent of* γ.

Proof. Fix $\gamma > 0$, and set

$$\Delta := \log\left(\frac{\gamma}{\tilde{\gamma}}\right).$$

For each $u \in (0, 1)$, let

$$t := \log u < 0, \qquad \delta_u := \delta + \frac{\Delta}{t}.$$

We first prove the key identity

$$E_\gamma(u; \delta) = E_{\tilde{\gamma}}(u; \delta_u).$$

Indeed, since $u = e^t$ and $e^\Delta = \gamma/\tilde{\gamma}$,

$$\tilde{\gamma} u^{\delta_u} = \tilde{\gamma} e^{(\delta + \Delta/t)t} = \tilde{\gamma} e^{\delta t} e^\Delta = \gamma u^\delta,$$

and similarly

$$\tilde{\gamma}^{-1} u^{2-\delta_u} = \tilde{\gamma}^{-1} e^{(2-\delta-\Delta/t)t} = \tilde{\gamma}^{-1} e^{(2-\delta)t} e^{-\Delta} = \gamma^{-1} u^{2-\delta}.$$

Substituting these identities into the definition of $E_{\tilde{\gamma}}$ yields

$$E_{\tilde{\gamma}}(u;\delta_u) = \frac{\log C\big(\tilde{\gamma}u^{\delta_u}, \tilde{\gamma}^{-1}u^{2-\delta_u}\big)}{\log u} = \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1}u^{2-\delta}\big)}{\log u} = E_{\gamma}(u;\delta).$$

We now prove that

$$\lim_{u\downarrow 0} E_{\gamma}(u;\delta) = \tilde{\kappa}(\delta).$$

Let $\varepsilon > 0$ be arbitrary. We must show that there exists $u^* \in (0,1)$ such that

$$\big|E_{\gamma}(u;\delta) - \tilde{\kappa}(\delta)\big| \le \varepsilon \qquad \text{for all } u \in (0,u^*].$$

Using the identity just established, it is enough to control

$$\big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta)\big|.$$

Add and subtract $\tilde{\kappa}(\delta_u)$, and apply the triangle inequality:

$$\big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta)\big| \le \big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta_u)\big| + \big|\tilde{\kappa}(\delta_u) - \tilde{\kappa}(\delta)\big|. \tag{1}$$

Alternatively, if one wishes to compare directly with $E_{\tilde{\gamma}}(u;\delta)$, then one may write

$$\big|E_{\tilde{\gamma}}(u;\delta_u) - E_{\tilde{\gamma}}(u;\delta)\big| \le \big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta_u)\big| + \big|\tilde{\kappa}(\delta_u) - \tilde{\kappa}(\delta)\big| + \big|\tilde{\kappa}(\delta) - E_{\tilde{\gamma}}(u;\delta)\big|. \tag{2}$$

We now control the relevant terms.

Step 1: control of $\delta_u - \delta$. Since Δ is fixed and $t = \log u \to -\infty$ as $u \downarrow 0$,

$$\delta_u - \delta = \frac{\Delta}{\log u} \longrightarrow 0.$$

Therefore, for every $r > 0$, there exists $u_1 \in (0,1)$ such that

$$|\delta_u - \delta| \le r \qquad \text{for all } u \in (0,u_1].$$

Step 2: control of $\big|\tilde{\kappa}(\delta_u) - \tilde{\kappa}(\delta)\big|$. By continuity of $\tilde{\kappa}$ at δ, there exists $r = r(\varepsilon) > 0$ such that

$$|d - \delta| \le r \quad \Longrightarrow \quad |\tilde{\kappa}(d) - \tilde{\kappa}(\delta)| \le \frac{\varepsilon}{2}.$$

By Step 1, there exists $u_2 \in (0,1)$ such that

$$|\delta_u - \delta| \le r \qquad \text{for all } u \in (0,u_2].$$

Hence,

$$|\tilde{\kappa}(\delta_u) - \tilde{\kappa}(\delta)| \le \frac{\varepsilon}{2} \qquad \text{for all } u \in (0, u_2]. \tag{3}$$

Step 3: control of $\big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta_u)\big|$. By local uniform convergence, there exist $r' = r'(\varepsilon) > 0$ and $u_3 = u_3(\varepsilon) \in (0,1)$ such that

$$|E_{\tilde{\gamma}}(u;d) - \tilde{\kappa}(d)| \le \frac{\varepsilon}{2}$$

for all $u \in (0, u_3]$ and all d satisfying $|d - \delta| \le r'$. Again by Step 1, there exists $u_4 \in (0,1)$ such that

$$|\delta_u - \delta| \le r' \qquad \text{for all } u \in (0, u_4].$$

Therefore,

$$|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta_u)| \le \frac{\varepsilon}{2} \qquad \text{for all } u \in (0, \min\{u_3, u_4\}]. \tag{4}$$

Step 4: conclusion. Let

$$u^* := \min\{u_2, u_3, u_4\}.$$

Then for all $u \in (0, u^*]$, combining (1), (3), and (4), we obtain

$$\big|E_{\tilde{\gamma}}(u;\delta_u) - \tilde{\kappa}(\delta)\big| \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $E_{\gamma}(u;\delta) = E_{\tilde{\gamma}}(u;\delta_u)$, it follows that

$$\big|E_{\gamma}(u;\delta) - \tilde{\kappa}(\delta)\big| \le \varepsilon \qquad \text{for all } u \in (0, u^*].$$

This proves that

$$\lim_{u \downarrow 0} E_{\gamma}(u;\delta) = \tilde{\kappa}(\delta).$$

Equivalently,

$$\kappa_L(\delta, \gamma; C) = \tilde{\kappa}(\delta) = \kappa_L(\delta, \tilde{\gamma}; C).$$

The proof is complete. □

Remark 2. *If one prefers to prove directly that*

$$\lim_{u \downarrow 0} \big(E_{\gamma}(u;\delta) - E_{\tilde{\gamma}}(u;\delta)\big) = 0,$$

then one may use (2) *instead of* (1)*. The three terms on the right-hand side are handled respectively by local uniform convergence, continuity of $\tilde{\kappa}$, and ordinary convergence of $E_{\tilde{\gamma}}(u;\delta) \to \tilde{\kappa}(\delta)$.*

Application to the Pareto-mixture copula

We now state the corresponding conclusion for the proposed copula C_{PM}.

Theorem 3. *Suppose that for the copula C_{PM}, the limit*

$$\kappa_L(\delta, \tilde{\gamma}; C_{\mathrm{PM}}) = \lim_{u \downarrow 0} \frac{\log C_{\mathrm{PM}}\big(\tilde{\gamma} u^{\delta}, \tilde{\gamma}^{-1} u^{2-\delta}\big)}{\log u}$$

exists for d in a neighborhood of δ, that the convergence is locally uniform in d, and that the resulting function $d \mapsto \kappa_L(d, \tilde{\gamma}; C_{\mathrm{PM}})$ is continuous at $d = \delta$. Then, for every $\gamma > 0$,

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \kappa_L(\delta, \tilde{\gamma}; C_{\mathrm{PM}}).$$

Hence $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ is independent of γ.

Proof. The proof is exactly the same as that of Theorem 1, with C replaced everywhere by C_{PM}. Indeed, define

$$E_{\gamma}^{\mathrm{PM}}(u; d) := \frac{\log C_{\mathrm{PM}}\big(\gamma u^{d}, \gamma^{-1} u^{2-d}\big)}{\log u}.$$

For fixed $\gamma > 0$, let

$$\Delta = \log\left(\frac{\gamma}{\tilde{\gamma}}\right), \qquad \delta_u = \delta + \frac{\Delta}{\log u}.$$

Then the same algebra gives

$$E_{\gamma}^{\mathrm{PM}}(u; \delta) = E_{\tilde{\gamma}}^{\mathrm{PM}}(u; \delta_u).$$

The same triangle inequality argument, together with local uniform convergence and continuity, yields

$$\lim_{u \downarrow 0} E_{\gamma}^{\mathrm{PM}}(u; \delta) = \kappa_L(\delta, \tilde{\gamma}; C_{\mathrm{PM}}).$$

Therefore,

$$\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \kappa_L(\delta, \tilde{\gamma}; C_{\mathrm{PM}}),$$

as claimed. □

Conclusion

Under the stated regularity conditions, the angular lower-tail coefficient

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u}$$

does not depend on the scaling parameter γ. The same conclusion holds for the proposed Pareto-mixture copula C_{PM}, provided the same existence, local uniformity, and continuity assumptions are satisfied.